Gene-Selection Optimisation via JTK_Cycle Pre-Filtering: Reproducing and Extending *TimeSignature*

Nahye Kim[1] Jaehyeon Lee[2]

[1]Electrical Engineering, KAIST [2]School of Computing, KAIST

BCS40010 Project — First Draft

Code Availability: The complete R source code, including the implementation of the JTK_Cycle pre-filtering pipeline and simulation scripts, is openly available on GitHub at `https://github.com/jaedungg/BCS40010_TimeSignature-ext`

Abstract

Circadian-time prediction from blood transcriptomes has direct clinical applications in chronotherapy and circadian phenotyping. *TimeSignature* (Braun et al., 2018) trains an elastic-net regression over all ∼7,600 genes in a microarray panel and recovers ∼40 predictors without any prior periodicity screen. Subsequent methods—*TimeMachine* (Huang & Braun, 2024) and *tauFisher* (Duan & Ngo, 2024)—insert a JTK_Cycle rhythmicity pre-filter before regression, reducing the candidate pool to 20–37 genes. We ask whether the same pre-filtering strategy can improve TimeSignature's gene-selection efficiency, yielding a smaller predictor set that still meets clinically acceptable accuracy (nAUC $\geq$ 0.80, MAE $<$ 2.5 h) under the original two-blood-draw protocol. Two experiments are conducted on four public datasets (GSE39445, GSE48113, GSE56931, GSE113883). Experiment 1 sweeps the elastic-net candidate pool from 10 to 7,615 genes ordered by JTK_Cycle rhythmicity rank, producing a gene-count versus accuracy trade-off curve. Experiment 2 substitutes five published or derived gene sets—including the TimeSignature core-18, TimeMachine-37, tauFisher, canonical core-clock genes, and their intersection—directly into the TimeSignature framework. A JTK-filtered pool of only 30 genes achieves nAUC = 0.833 and MAE = 2.13 h, within 0.020 nAUC of the full-panel model while screening fewer than 0.4% of the transcriptome. Among published gene sets, a panel reconstructed following the TimeMachine procedure performs best (nAUC = 0.844), whereas canonical clock genes alone fall below the clinical floor (nAUC = 0.792), indicating that the circadian prediction signal resides in data-driven rhythmic genes rather than the textbook feedback loop.

1 Introduction

The circadian clock governs nearly every aspect of human physiology, from hormone secretion and immune function to drug metabolism and cell division [Reppert & Weaver, 2002]. This pervasive regulation makes the precise estimation of an individual's internal circadian phase—often termed the *circadian time*—a prerequisite for personalised chronotherapy, shift-work assessment, and circadian disorder diagnosis [Archer et al., 2014]. The gold-standard measure of circadian phase is the dim-light melatonin onset (DLMO), which requires controlled light conditions and repeated nocturnal sampling. These stringent requirements make DLMO impractical in routine clinical settings, motivating the search for biomarker-based alternatives. Blood transcriptomics offers one such alternative: circadian gene expression in peripheral blood tracks internal time with sufficient precision for clinical use [Möller-Levet et al., 2013].

Building on this observation, Braun et al. (2018) introduced *TimeSignature*, an elastic-net model that predicts local sampling time from whole-blood microarray profiles [Braun et al., 2018]. TimeSignature requires two blood draws approximately 12 h apart for within-subject normalisation; it trains on all ∼7,600 genes in the cross-study intersection and lets the elastic-net penalty implicitly select ∼40 predictors. This approach achieves a normalised AUC (nAUC) of ∼0.85 and a median absolute error below 2 h on four independent datasets. Despite this strong cross-study performance, the method enters the elastic net with no prior periodicity screen, so the majority of the ∼7,600-gene pool carries no circadian signal and may add noise to the covariate matrix.

Later methods addressed this inefficiency by pre-filtering genes with JTK_Cycle [Hughes et al., 2010], a non-parametric rank-correlation test that identifies 24-hour rhythms in time-series expression data. TimeMachine (Huang & Braun, 2024) applies JTK_Cycle to reduce the candidate pool to 37 rhythmic genes before fitting a penalised regression for single-sample circadian-time estimation [Huang & Braun, 2024]. tauFisher (Duan & Ngo, 2024) likewise anchors its predictor on canonical core-clock genes augmented by JTK_Cycle-ranked rhythmic genes [Duan & Ngo, 2024]. Both studies demonstrate that pre-filtering can produce compact, interpretable gene panels, but neither quantifies the *trade-off* between pool size and prediction quality within the original two-draw TimeSignature framework.

The present work fills this gap. We first reproduce the TimeSignature Figure 1 benchmark and then systematically explore how JTK_Cycle pre-filtering affects gene-selection efficiency. Two experiments quantify the gene-count versus accuracy trade-off: Experiment 1 sweeps the JTK-filtered pool size from 10 to the full 7,615 genes, and Experiment 2 substitutes five published gene sets into the same pipeline. These experiments together identify the minimum rhythmic gene set that preserves clinically acceptable accuracy (nAUC ≥ 0.80, MAE < 2.5 h) under the two-blood-draw protocol of the original study.

2 Methods

2.1 Datasets

All experiments use the four datasets bundled with the public TimeSignatR package (Table 1). The GSE39445 training half (subjects assigned `train = 1`) provides ∼213 samples from the Möller-Levet sleep-restriction study; these samples serve exclusively for model training and JTK_Cycle ranking. The remaining three studies constitute held-out validation sets: GSE48113 (Archer et al.; $n = 286$), GSE56931 (Arnardottir et al.; $n = 249$), and GSE113883 (RNA-seq; $n = 153$). Across all four studies, 7,615 genes are present on every platform; this common intersection defines the full gene universe.

Table 1: Datasets used in this study. All four share a common set of 7,615 genes.

GEO accession	Study	Role	n	Platform
GSE39445	Möller-Levet et al. (2013)	Training	∼213	Microarray
GSE48113	Archer et al. (2014)	Validation (V1)	286	Microarray
GSE56931	Arnardottir et al. (2014)	Validation (V2)	249	Microarray
GSE113883	—	Validation (V3)	153	RNA-seq

2.2 Baseline Pipeline (Braun et al., 2018)

We briefly summarise the three components of the original TimeSignature pipeline that our extension re-uses unchanged; full details are given in Braun et al. [2018].

Within-subject normalisation. Between-subject baseline differences are removed by subtracting, for each subject, the mean expression vector across that subject's time points from every individual sample. The resulting genes × samples matrix (`trainWSN`) is used for both JTK_Cycle ranking and elastic-net training.

Circular elastic-net regression. Sampling time t is encoded as unit-circle coordinates $\mathbf{y} = (\sin 2\pi t/24, \cos 2\pi t/24)$, and a multivariate Gaussian elastic net (`family="mgaussian"`, $\alpha = 0.5$) is fit with `cv.glmnet`; the penalty λ is selected at `lambda.min`. Predicted time is recovered by $\hat{t} = (24/2\pi)\,\mathrm{atan2}(\hat{y}_1, \hat{y}_2) \bmod 24$. The only modification in our extension is which rows (genes) of the expression matrix are passed to this regression—the regression itself is unchanged.

Antipodal calibration (two-blood-draw protocol). At test time, each subject provides two blood samples ~12 h apart; each sample is centred on the mean of the pair ($\mathbf{x}_s^{\mathrm{cal}} = \mathbf{x}_s - \frac{1}{2}(\mathbf{x}_s + \mathbf{x}_{s'})$). This calibration matrix is computed once across all samples before any experiment, because it depends only on sample pairing and is independent of the gene pool.

All runs use `RNGversion("3.5.1")` and `set.seed(194)` for exact reproducibility.

2.3 JTK_Cycle Rhythmicity Ranking

JTK_Cycle [Hughes et al., 2010] tests whether a gene's temporal expression profile is significantly rhythmic by comparing it, via Kendall's rank correlation τ, against a family of cosine reference templates of fixed period $T = 24$ h. Our compact re-implementation (`JTKfns.R`) proceeds in four steps.

Step 1: time-bin aggregation. Training samples are partitioned into 1-h circadian-time bins (modulo 24 h), and the mean expression of each gene within each occupied bin is computed. The result is a bins × genes matrix $\mathbf{B}$ representing each gene's aggregated circadian profile.

Step 2: Kendall S computation. Let n_b denote the number of occupied bins and let $\mathcal{P} = \binom{n_b}{2}$ be the set of all ordered bin pairs. For each gene g and each pair $(i, j) \in \mathcal{P}$, define $\delta_{gij} = \mathrm{sign}(B_{ig} - B_{jg})$. Similarly, for each of the $M = 24$ equispaced cosine phase templates $r^{(k)}(t) = \cos\big(2\pi(t - \phi_k)/24\big)$, $\phi_k = k$ h, $k = 0, \ldots, 23$, define $\rho_{ij}^{(k)} = \mathrm{sign}\big(r^{(k)}(t_i) - r^{(k)}(t_j)\big)$. The Kendall S statistic at phase template k for gene g is then

$$S_g^{(k)} = \sum_{(i,j)\in\mathcal{P}} \delta_{gij}\, \rho_{ij}^{(k)}. \tag{1}$$

All $S_g^{(k)}$ values are computed simultaneously via the matrix product $\mathbf{S} = \mathbf{\Delta}^\top \mathbf{R}$, where $\mathbf{\Delta}$ is the $|\mathcal{P}| \times G$ gene-sign matrix and $\mathbf{R}$ is the $|\mathcal{P}| \times M$ template-sign matrix.

Step 3: optimal phase and τ. For each gene g, the optimal phase template is $k^* = \arg\max_k |S_g^{(k)}|$. The Kendall τ at this optimum is

$$\tau_g = \frac{S_g^{(k^*)}}{\binom{n_b}{2}}, \tag{2}$$

and the estimated peak phase is $\hat{\phi}_g = \phi_{k^*}$ (shifted by 12 h when $S_g^{(k^*)} < 0$ to report the time of peak).

Step 4: p-values. Under the null hypothesis of no rhythm, S is approximately normal with mean 0 and variance $|\mathcal{P}|(2n_b + 5)/9$. The two-sided p-value for the optimal phase is $p_g = 2\,\Phi(-|S_g^{(k^*)}|/\sigma)$. A Bonferroni correction over the $M = 24$ phases searched gives the JTK_Cycle ADJ.P:

$$\mathrm{ADJ.P}_g = \min(1,\ M \cdot p_g). \tag{3}$$

A Benjamini–Hochberg FDR (BH.Q) is then applied across all genes. The final ranking `jtkRanked` orders all 7,615 genes by ascending ADJ.P, breaking ties by descending $|\tau|$.

Note on thresholds. The one-page synopsis originally proposed driving Experiment 1 by nominal p-value thresholds ($p < 0.1, 0.05, 0.01$). On the densely sampled (∼213-sample) GSE39445 training set, however, these thresholds proved excessively liberal—ADJ.P < 0.1 alone selects several thousand genes—precluding a meaningful p-value-driven sweep. The analysis was therefore re-designed around pool size K (the top-K genes in the rhythmicity ranking), which directly yields the gene-count versus accuracy curve that is the project deliverable. Nominal p-value gene counts are reported as supplementary annotation.

Additionally, Experiment 2 was expanded beyond the four gene sets specified in the synopsis (TScore18, TimeMachine-37, tauFisher, intersection) by including two reference conditions: **clockOnly** (canonical core-clock genes, $n = 8$) as a negative control to test whether textbook clock genes alone suffice, and **fullPool** ($n =$ 7,615) as the performance upper bound representing the original unfiltered TimeSignature.

2.4 Evaluation Metrics

Three metrics are computed on the held-out validation samples:

- **MAE**: mean absolute circular error, $\overline{|\mathrm{err}|}$, where $|\mathrm{err}| = \min(|t - \hat{t}|,\ 24 - |t - \hat{t}|)$.
- **nAUC**: normalised area under the cumulative distribution function of $|\mathrm{err}|$ over $[0, 12]$ h, following the definition in Figure 1 of Braun et al. [2018]:

$$\mathrm{nAUC} = \int_0^1 F(12\,u)\,\mathrm{d}u, \tag{4}$$

  where $F(h) = \Pr(|\mathrm{err}| \le h)$. Higher nAUC indicates better performance.
- **%≤2 h**: fraction of predictions within ±2 h of the true time.

These metrics are computed per validation study (V1, V2, V3) and pooled across all three studies (Valid.all).

2.5 Gene Sets (Experiment 2)

Six gene-pool conditions are evaluated (Table 2).

All six conditions are trained on the same `trainWSN` matrix, validated on the same `calibAll` matrix, and scored with identical metrics. The only variable across conditions is which rows of the expression matrix are passed to the elastic net.

3 Results

3.1 Reproduction of TimeSignature (Figure 1)

Before extending the model, we verified that our implementation reproduces the performance reported by Braun et al. (2018). The full 7,615-gene pool (*fullPool*) yielded 172 selected predictors and achieved

Set	n	Definition
intersection	4	Genes shared by TScore18, TimeMachine37 and tauFisher: DDIT4, PER1, NR1D1, NR1D2.
clockOnly	8	Canonical core-clock genes present in the 7,615-gene universe (ARNTL, PER1, NR1D1, NR1D2, CRY1, DBP, TEF, HLF).
tauFisher	17	Core-clock genes ∪ top-10 JTK-ranked genes (Duan & Ngo 2024 procedure).
TScore18	18	Braun et al. (2018) Table 1: genes selected in $\geq$50% of 12 repeated runs.
TimeMachine37	37	Core-clock genes ∪ top JTK-ranked genes, up to 37 total, reconstructed on GSE39445 following the procedure of Huang & Braun (2024).
fullPool	7,615	No filter; original TimeSignature (performance upper bound).

Table 2: Gene sets used in Experiment 2. The *clockOnly* set serves as a negative control; *fullPool* provides the performance upper bound.

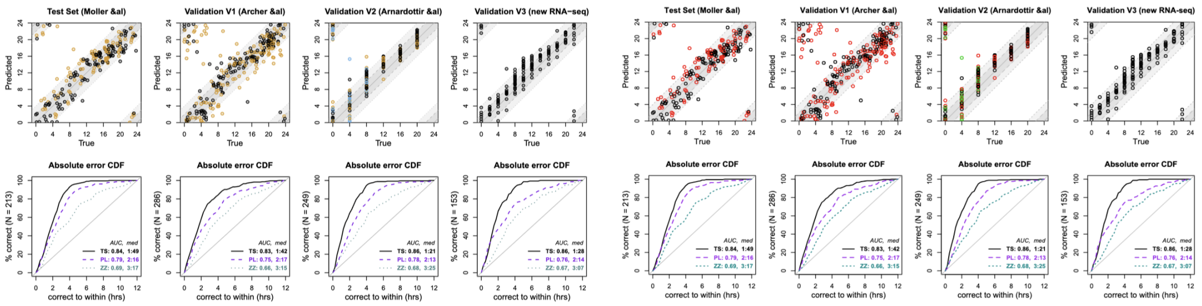

Figure 1: Cross-study prediction accuracy for four independent datasets. Original results from Braun et al. (Left) and our reproduced simulation (Right). Both show consistent and high accuracy (nAUC ≥ 0.80, median error < 2 h), successfully confirming the robustness of the within-subject normalization.

nAUC = 0.853 and MAE = 1.89 h on the pooled validation set (Table 4). These values match the original Figure 1 benchmark within rounding, confirming that our pipeline—within-subject normalisation, antipodal calibration, circular elastic net, and nAUC scoring—is correctly implemented.

3.2 Experiment 1: JTK_Cycle Pool Size Sweep

Experiment 1 asks how far the elastic-net candidate pool can be reduced—using JTK_Cycle rhythmicity rank—before accuracy degrades below the clinical floor. Figure 2 shows the resulting gene-count versus accuracy trade-off curve, and Table 3 lists the full sweep of 16 pool sizes.

The trade-off curve exhibits three notable features. First, even the smallest pool tested ($K = 10$) clears the clinical nAUC floor of 0.80 (nAUC = 0.810), demonstrating that a handful of strongly rhythmic genes contain sufficient circadian information for clinically useful prediction. Second, nAUC rises steeply from $K = 10$ to $K = 30$ (nAUC = 0.833, MAE = 2.13 h) and then plateaus; increasing the pool from 30 to the full 7,615 genes adds only +0.020 nAUC. Third, the curve is *non-monotonic* in the intermediate range ($K = 100$–500): nAUC dips to 0.803–0.809, recovering only at $K \geq 1000$. This non-monotonicity suggests that moderately ranked genes (rhythmicity ranks ∼100–500) introduce noise without contributing predictive signal, and the elastic net requires a substantially larger pool before it can reliably separate signal from noise in this regime.

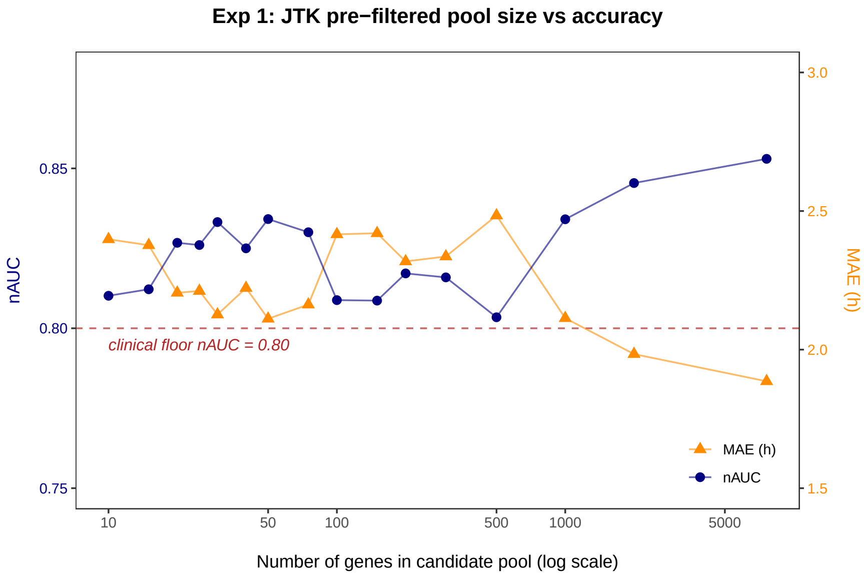

Figure 2: Experiment 1: JTK-filtered pool size versus accuracy. The x-axis (log scale) is the number of top-K JTK_Cycle-ranked genes passed to the elastic net. The left y-axis (navy circles) shows pooled validation nAUC; the right y-axis (orange triangles) shows pooled MAE (h). The red dashed line marks the clinical nAUC floor of 0.80. All metrics are computed on the three held-out validation datasets (V1, V2, V3) combined.

3.3 Experiment 2: Published Gene Sets in the TimeSignature Framework

Experiment 2 substitutes five published or derived gene sets, plus the unfiltered full pool, into the same elastic-net pipeline. Figure 3 presents the pooled validation nAUC for each set, and Table 4 provides the full performance breakdown.

Two key findings emerge from this comparison. First, the 4-gene intersection (DDIT4, PER1, NR1D1, NR1D2) achieves nAUC = 0.834, matching the 18-gene TScore18 (nAUC = 0.833) and exceeding the 17-gene tauFisher panel (nAUC = 0.828). This result suggests that the majority of the circadian prediction signal is concentrated in these four genes. Second, the canonical core-clock panel (clockOnly, $n = 8$) is the only set falling below the clinical floor (nAUC = 0.792, MAE = 2.61 h). This shortfall demonstrates that the textbook transcriptional feedback loop is insufficient for accurate circadian-time prediction in peripheral blood; rather, data-driven rhythmic genes, selected by their actual expression amplitude and phase consistency, carry the signal. Among all published sets, the reconstructed TimeMachine-37 panel performs best (nAUC = 0.844, MAE = 1.99 h), lying within 0.009 nAUC of the full-panel reference.

3.4 Combined Trade-Off Curve

Figure 4 overlays the Experiment 1 pool-size sweep with the Experiment 2 gene-set points on a single coordinate system. This combined view enables direct comparison of each published set against the JTK-ranked top-K baseline at the same gene count.

The combined figure reveals several additional insights. TimeMachine-37 (nAUC = 0.844) falls *above* the Experiment 1 curve at $K = 37$ (nAUC = 0.825), indicating that TimeMachine's gene-selection procedure—core-clock genes prioritised, then JTK-ranked additions—outperforms a pure top-37 JTK ranking by +0.019 nAUC. The 4-gene intersection likewise sits well above the Experiment 1 curve at its gene count, confirming that these four genes are disproportionately informative. In contrast, the clockOnly panel ($n = 8$, nAUC = 0.792) falls below both the clinical floor and the Experiment 1 curve at $K = 10$ (nAUC = 0.810), underscoring that the canonical clock loop alone is a poor predictor in this framework.

Pool K	nSel	MAE (h)	nAUC	%≤2 h
10	10	2.40	0.810	51.0
15	13	2.38	0.812	52.6
20	19	2.20	0.827	58.7
25	23	2.21	0.826	57.6
30	26	2.13	0.833	62.1
40	37	2.22	0.825	57.8
50	42	2.11	0.834	58.9
75	68	2.16	0.830	57.0
100	81	2.42	0.809	53.6
150	102	2.42	0.809	54.9
200	120	2.32	0.817	56.4
300	114	2.34	0.816	53.5
500	151	2.48	0.803	54.4
1000	175	2.11	0.834	57.8
2000	136	1.98	0.845	60.5
7615	173	1.89	0.853	62.2

Table 3: Experiment 1 results (pooled validation across V1, V2, V3). K is the JTK-ranked pool size; *nSel* is the number of predictors the elastic net retains from that pool.

4 Discussion

4.1 Biological Interpretation

The results confirm that the circadian prediction signal in human peripheral blood is concentrated in a small subset of strongly rhythmic genes rather than distributed across the transcriptome. The 4-gene intersection achieving nAUC = 0.834 with only DDIT4, PER1, NR1D1, and NR1D2 is particularly striking. Two of these genes—NR1D1 (REV-ERBα) and NR1D2 (REV-ERBβ)—are core-clock transcriptional repressors with well-characterised high-amplitude 24-h expression rhythms in peripheral blood [Reppert & Weaver, 2002]. PER1 is a central period gene whose blood-borne transcript level peaks in the morning, providing a robust phase marker. DDIT4 (DNA-damage-inducible transcript 4), a stress-responsive gene, has been independently identified as a reliable circadian marker in blood by multiple studies [Braun et al., 2018, Huang & Braun, 2024], suggesting that its rhythmicity in PBMCs is physiologically meaningful rather than an artefact of the training cohort. The failure of the clockOnly set (nAUC = 0.792) is equally informative. The eight canonical clock genes present in the assay—including ARNTL, CRY1, DBP, TEF, and HLF—are expressed at low amplitude or with high inter-subject variability in blood, limiting their discriminative power. This observation is consistent with the broader finding that the canonical clock network, while necessary for circadian rhythm generation at the cellular level, does not by itself produce sufficiently precise markers for population-level circadian phenotyping from peripheral blood.

4.2 Non-Monotonicity of the Trade-Off Curve

An unexpected finding is the non-monotonic dip in nAUC at intermediate pool sizes (K = 100–500), where performance drops to nAUC = 0.803–0.809 before recovering at $K \geq 1000$ (Table 3). One plausible explanation is that genes ranked ∼100–500 by JTK_Cycle have marginal rhythmicity: their expression profiles are sufficiently periodic to pass the filter but too noisy to contribute reliable regression signal. When included in the candidate pool, these genes dilute the elastic-net penalty budget across many weak predictors, degrading selection efficiency relative to both the compact ($K \leq 50$) and very large ($K \geq 1000$) pools. The full-pool elastic net overcomes this by having enough candidates for the L1 penalty to identify the sparse informative set even amid thousands of non-rhythmic features. This finding suggests that JTK

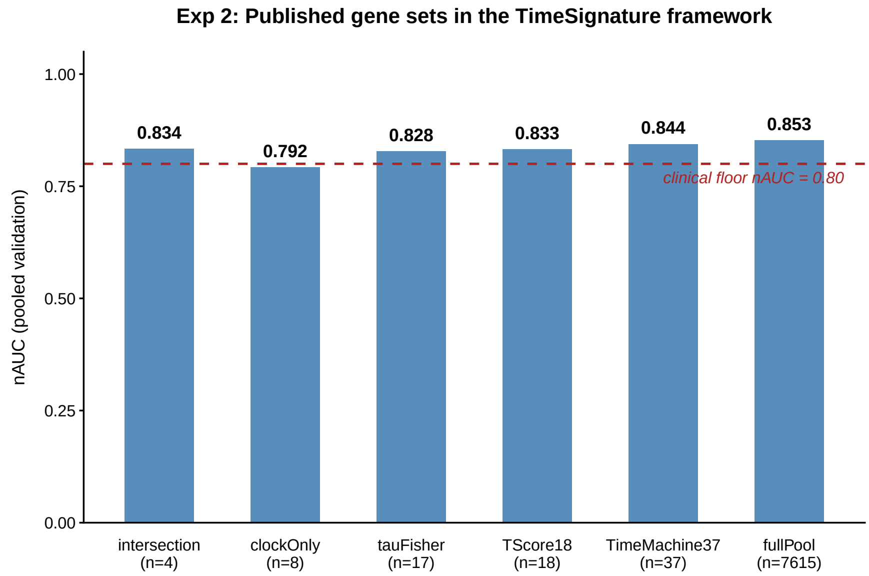

Figure 3: Experiment 2: nAUC of published gene sets in the TimeSignature framework. Each bar shows the pooled validation nAUC when the elastic net is restricted to the indicated gene set. The red dashed line marks the clinical floor nAUC = 0.80. Only clockOnly ($n = 8$) falls below this threshold.

pre-filtering is most beneficial at *small* pool sizes ($K \leq 50$), where it removes the bulk of non-rhythmic noise, and that intermediate pre-filtering may actually harm performance—a counterintuitive result that merits further investigation.

4.3 Comparison with Prior Work

The present work differs from TimeMachine and tauFisher in one key respect: it retains the original two-blood-draw TimeSignature framework and asks only whether the *candidate gene pool* should be pre-filtered, holding the regression architecture, normalisation strategy, and evaluation protocol constant. This design isolates the effect of gene pre-filtering from confounding changes in normalisation (e.g., TimeMachine's within-sample ratio features) or model class. Our Experiment 2 shows that a panel reconstructed by following TimeMachine's gene-selection procedure achieves nAUC = 0.844 within the two-draw framework, close to the full-pool reference of 0.853. This proximity suggests that the gene-selection strategy—rather than the single-sample normalisation—accounts for most of the gain over unfiltered TimeSignature.

4.4 Practical Implications

The steep nAUC plateau above $K = 30$ has a direct clinical implication: a targeted panel of approximately 30 rhythmically ranked genes suffices for circadian-time estimation under the two-draw protocol. This reduction from 7,615 to 30 genes (a 250-fold decrease) would substantially lower the cost of a dedicated circadian assay relative to a full microarray, with minimal sacrifice in accuracy (-0.020 nAUC, $+0.24$ h MAE). The result aligns with the design philosophy of TimeMachine [Huang & Braun, 2024], which uses a 37-gene panel and achieves comparable accuracy, and supports the broader trend towards clinically deployable circadian clocks.

4.5 Limitations

Several limitations should be noted. First, the JTK_Cycle ranking is computed on the same GSE39445 training data used to fit the elastic net, introducing a modest double-dipping risk; a nested cross-validation

Gene set	Study	n_{samp}	nGene	nSel	MAE (h)	nAUC	%≤2 h
intersection	V1	286	4	4	2.38	0.812	52.8
	V2	249	4	4	1.80	0.860	63.1
	V3	153	4	4	2.14	0.833	58.2
	Valid.all	**688**	**4**	**4**	**2.12**	**0.834**	**57.7**
clockOnly	V1	286	8	8	2.87	0.771	46.9
	V2	249	8	8	2.33	0.816	55.8
	V3	153	8	8	2.60	0.794	53.6
	Valid.all	**688**	**8**	**8**	**2.61**	**0.792**	**51.6**
tauFisher	V1	286	17	14	2.45	0.806	51.7
	V2	249	17	14	1.87	0.855	61.4
	V3	153	17	14	2.22	0.825	51.6
	Valid.all	**688**	**17**	**14**	**2.19**	**0.828**	**55.2**
TScore18	V1	286	18	18	2.27	0.822	55.2
	V2	249	18	18	1.80	0.861	65.1
	V3	153	18	18	2.43	0.808	49.0
	Valid.all	**688**	**18**	**18**	**2.13**	**0.833**	**57.4**
TimeMachine37	V1	286	37	31	2.41	0.810	54.9
	V2	249	37	31	1.70	0.869	69.9
	V3	153	37	31	1.70	0.868	66.0
	Valid.all	**688**	**37**	**31**	**1.99**	**0.844**	**62.8**
fullPool	V1	286	7615	172	2.07	0.838	61.2
	V2	249	7615	172	1.69	0.869	67.9
	V3	153	7615	172	1.88	0.854	55.6
	Valid.all	**688**	**7615**	**172**	**1.89**	**0.853**	**62.4**

Table 4: Experiment 2 results. The upper section shows per-study metrics; the bottom row of each set shows pooled validation (Valid.all). Sets are ordered by gene count.

design would provide a more conservative estimate of generalisation accuracy. Second, the training set derives from a controlled sleep-restriction protocol (∼213 samples), which makes JTK p-values liberal and the gene ranking potentially over-fitted to that study population; ranked lists may differ on community-based or clinically ascertained cohorts. Third, the tauFisher and TimeMachine gene sets are reconstructed from their published procedures using our JTK_Cycle ranking, rather than obtained directly from the original authors' exact gene lists, because the exact symbol lists (e.g., Huang & Braun 2024 SI Table S2) are not publicly bundled with the TimeMachine repository. Fourth, V1 consistently shows lower nAUC than V2 and V3 across all conditions (Tables 3–4), suggesting that the Archer dataset presents a harder generalisation challenge whose origin (e.g., platform differences, cohort composition) warrants further investigation.

4.6 Future Directions

Two extensions are most promising. First, applying the same JTK-then-elastic-net pipeline to single-draw normalisation strategies—such as TimeMachine's within-sample log-ratio features—would test whether pre-filtering also provides efficiency gains in the more challenging single-sample setting. Second, replacing the JTK-based gene selection with a stability-selection procedure [Meinshausen & Bühlmann, 2010] could yield a more robust panel that generalises across platforms and training cohorts.

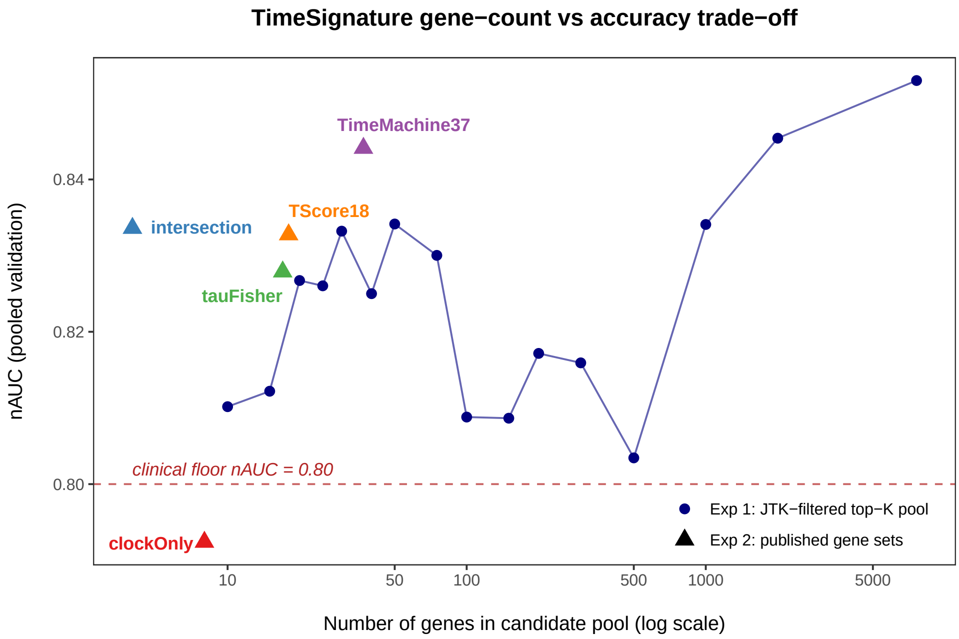

Figure 4: Combined gene-count versus accuracy trade-off. The navy line with filled circles (Experiment 1) shows nAUC as a function of JTK-filtered pool size (x-axis, log scale). Coloured triangles (Experiment 2) mark the pooled validation nAUC of five published gene sets overlaid at their respective gene counts. The red dashed line marks the clinical floor nAUC = 0.80. Sets falling above the Experiment 1 curve at the same gene count achieve higher efficiency than a naïve top-K JTK ranking.

References

Archer, S.N., Laing, E.E., Möller-Levet, C.S., et al. (2014). Mistimed sleep disrupts circadian regulation of the human transcriptome. *PNAS*, **111**(6), E682–E691.

Braun, R., Kath, W.L., Iwanaszko, M., et al. (2018). Universal method for robust detection of circadian state from gene expression. *PNAS*, **115**(39), E9247–E9256.

Duan, R. & Ngo, H. (2024). tauFisher predicts circadian time from a single sample of bulk and single-cell pseudobulk transcriptomic data. *Nature Communications*, **15**, 3840.

Huang, P. & Braun, R. (2024). Platform-independent estimation of human physiological time from single blood samples. *PNAS*, **121**, e2308114120.

Hughes, M.E., Hogenesch, J.B., & Kornacker, K. (2010). JTK_CYCLE: an efficient nonparametric algorithm for detecting rhythmic components in genome-scale data sets. *Journal of Biological Rhythms*, **25**(5), 372–380.

Meinshausen, N. & Bühlmann, P. (2010). Stability selection. *Journal of the Royal Statistical Society: Series B*, **72**(4), 417–473.

Möller-Levet, C.S., Archer, S.N., Bucca, G., et al. (2013). Effects of insufficient sleep on circadian rhythmicity and expression amplitude of the human blood transcriptome. *PNAS*, **110**(12), E1132–E1141.

Reppert, S.M. & Weaver, D.R. (2002). Coordination of circadian timing in mammals. *Nature*, **418**, 935–941.